

Singular Point

- Recall that a point $x = x_0$ is called the **regular** or **ordinary** point of the second order homogeneous differential equation

$$y'' + p(x)y' + q(x)y = 0$$

if $p(x)$ and $q(x)$ are analytic at $x = x_0$.

- If $x = x_0$ is not regular point then it is called a **singular point**
- A singular point is further classified as
 - regular singular point
 - irregular singular point

Regular Singular Point

The point $x = x_0$ is a *regular singular point* of the equation

$$y'' + p(x)y' + q(x)y = 0$$

if $x = x_0$ is not an ordinary point but both $(x - x_0)p(x)$ and $(x - x_0)^2q(x)$ are analytic at $x = x_0$.

Example 1.1

- Consider

$$2x^2y'' + 7x(x+1)y' - 3y = 0.$$

Here, the point $x = 0$ is a regular singular point

- Consider

$$8x^2y'' + 10xy' + (x-1)y = 0.$$

Here, the point $x = 0$ is a regular singular point

Method of Frobenius

If $x = 0$ is a regular singular point of

$$y'' + p(x)y' + q(x)y = 0, \quad (2.1)$$

then this equation has at least one solution of the form

$$y = x^r \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^{n+r}, \quad (2.2)$$

where r and a_n ($n = 0, 1, 2, \dots$) are constants. This series is known as the **Frobenius series**, which is convergent in the interval $0 < x < R$ for some real R . Then

$$y' = \sum_{n=0}^{\infty} (n+r)a_n x^{n+r-1}, \text{ and } y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{n+r-2} \quad (2.3)$$

Insert (2.2) and (2.3) into (2.1).

Computing the Coefficients a_n and r in (2.2)

- Collect the coefficients of like powers of x and equate to zero
- With the coefficient of general power x^{n+r} , the resulting relation will give a recurrence formula
- Equating to zero the coefficient of the lowest power of x , we obtain a quadratic equation in r , called the **indicial equation**, of the problem
- Find the two roots r_1 and r_2 of the indicial equation. Suppose that r_1 and r_2 do not differ by an integer.
- Corresponding to each **indicial root**, use the recurrence relation to determine the respective coefficients, and hence two linearly independent Frobenius series solutions $y_1(x)$ and $y_2(x)$ (2.1).
- Its general solution is $y = c_1 y_1(x) + c_2 y_2(x)$, where c_1 and c_2 are arbitrary constants

Example 2.1

Find a series solution near $x = 0$ for

$$3x^2y'' - xy' + y = 0 \text{ or } y'' - \frac{1}{3x}y' + \frac{1}{3x^2}y = 0. \quad (2.4)$$

Solution.

Comparing (2.4) with the standard form

$$y'' + p(x)y' + q(x)y = 0,$$

we have

$$p(x) = -1/3x \text{ and } Q(x) = 1/3x^2,$$

which are not defined at $x = 0$. But,

$$xP(x) = -1/3 \text{ and } x^2Q(x) = 1/3,$$

being constant functions, are analytic at $x = 0$. Therefore, $x = 0$ is a regular singular point for (2.4).

Now, we assume the Frobenius series solution $y = \sum_{n=0}^{\infty} a_n x^{n+r}$. Then

$$y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}, \text{ and } y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Insert all these into (2.4), we get

$$3x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} - x \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} + \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

or

$$\sum_{n=0}^{\infty} [3(n+r)(n+r-1) - (n+r) + 1] a_n x^{n+r} = 0$$

which is simplified as

Equating to zero, the coefficients of the lowest power of x , namely, that of x^r in (2.5), we obtain the indicial equation

$$3r^2 - 4r + 1 = 0. \quad (2.6)$$

The roots of (2.6) are $r_1 = 1$ and $r_2 = 1/3$, which do not differ by an integer.

Now equating to zero, the coefficient of x^{n+r} , $n \geq 1$, we get the recurrence relation:

$$[3(r+n)^2 - 4(r+n) + 1]a_n = 0.$$

Since $3(r+n)^2 - 4(r+n) + 1 \neq 0$ for $n \geq 1$, we should have

$$a_n = 0, n \geq 1.$$

That is,

$$a_1 = a_2 = \cdots = 0$$

With the values of a_n , the two solutions corresponding to the two roots $r_1 = 1$ and $r_2 = 1/3$, are finally given by

$$y_1(x) = x^{r_1} \sum_{n=0}^{\infty} a_n(r_1)x^n = a_0x \text{ and } y_2(x) = x^{r_2} \sum_{n=0}^{\infty} a_n(r_2)x^n = a_0x^{1/3}.$$

Therefore, the general solution of (2.4) is

$$y(x) = c_1y_1(x) + c_2y_2(x) = a_0 \left[c_1x + c_2x^{1/3} \right]$$

The complete solution depends on the nature of roots of the indicial equation.

Case I. When roots of the indicial equation are distinct and do not differ by an integer, the complete solution is

$$y = c_1(y)_{m_1} + c_2(y)_{m_2}$$

where m_1, m_2 are the roots.

Example:

Solve in series the equation $9x(1-x) \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 4y = 0$.

Solution. Here $x = 0$ is a singular point since coefficient of $y'' = 0$ at $x = 0$.

Substituting $y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots$

$$\therefore \frac{dy}{dx} = m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots$$

$$\frac{d^2 y}{dx^2} = m(m-1) a_0 x^{m-2} + (m+1) m a_1 x^{m-1} + (m+2)(m+1) a_2 x^m + \dots$$

in the given equation, we obtain

$$9x(1-x) [m(m-1)a_0x^{m-2} + (m+1)ma_1x^{m-1} + (m+2)(m+1)a_2x^m + \dots] \\ - 12[ma_0x^{m-1} + (m+1)a_1x^m + (m+2)a_2x^{m+1} + \dots] + 4[a_0x^m + a_1x^{m+1} + a_2x^{m+2} + \dots] = 0.$$

The lowest power of x is x^{m-1} . Its coefficient equated to zero gives

$$a_0(9m(m-1) - 12m) = 0, \text{ i.e., } m(3m-7) = 0 \quad \text{as } a_0 \neq 0.$$

Thus, the roots of the *indicial equation* are $m = 0, 7/3$.

i.e., Roots are distinct and do not differ by an integer.

The coefficient of x^m equated to zero gives

$$a_1\{9(m+1)m - 12(m+1)\} + a_0\{4 - 9m(m-1)\} = 0$$

$$3a_1(3m-4)(m+1) - a_0(3m-4)(3m+1) = 0$$

$$3a_1(m+1) = a_0(3m+1).$$

Similarly $3a_2(m+2) = a_1(3m+4)$, $3a_3(m+3) = a_2(5m+7)$ and so on.

$$\therefore a_1 = \frac{3m+1}{3(m+1)} a_0, a_2 = \frac{(3m+4)a_1}{3(m+2)} = \frac{(3m+4)(3m+1)}{3^2(m+2)(m+1)} a_0, a_3 = \frac{(3m+7)(3m+4)(3m+1)}{3^3(m+3)(m+2)(m+1)} a_0 \text{ etc.}$$

When $m = 0$, $a_1 = \frac{1}{3} a_0$, $a_2 = \frac{1.4}{3.6} a_0$, $a_3 = \frac{1.4.7}{3.6.9} a_0$ etc. giving the particular solution

$$y_1 = a_0 \left[1 + \frac{1}{3}x + \frac{1.4}{3.6}x^2 + \frac{1.4.7}{3.6.9}x^3 + \dots \right]$$

When $m = 7/3$, the particular solution is

$$y_2 = a_0 x^{7/3} \left[1 + \frac{8}{10}x + \frac{8.11}{10.13}x^2 + \frac{8.11.14}{10.13.16}x^3 + \dots \right]$$

Thus the complete solution is $y = c_1 y_1 + c_2 y_2$

$$y = C_1 \left[1 + \frac{1}{3}x + \frac{1 \cdot 4}{3 \cdot 6}x^2 + \frac{1 \cdot 4 \cdot 7}{3 \cdot 6 \cdot 9}x^3 + \dots \right]$$

$$+ C_2 x^{7/3} \left[1 + \frac{8}{10}x + \frac{8 \cdot 11}{10 \cdot 13}x^2 + \frac{8 \cdot 11 \cdot 14}{10 \cdot 13 \cdot 16}x^3 + \dots \right],$$

where $C_1 = c_1 a_0$, $C_2 = c_2 a_0$.

Case II. When roots of the indicial equation are equal *the complete solution is*

$$y = c_1(y)_{m_1} + c_2 \left(\frac{\partial y}{\partial m} \right)_{m_1}$$

where m_1, m_1 are the roots.

Example:

Solve in series the equation $x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + xy = 0$.

Solution. Substituting $y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots$

$$\frac{dy}{dx} = m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots$$

$$\frac{d^2 y}{dx^2} = m(m-1) a_0 x^{m-2} + (m+1)m a_1 x^{m-1} + (m+2)(m+1) a_2 x^m + \dots$$

in the given equation, we obtain

$$\begin{aligned} & x[m(m-1)a_0x^{m-2} + (m+1)ma_1x^{m-1} + (m+2)(m+1)a_2x^m + \dots] \\ & + [ma_0x^{m-1} + (m+1)a_1x^m + (m+2)a_2x^{m+1} + \dots] \\ & + x[a_0x^m + a_1x^{m+1} + a_2x^{m+2} + \dots] = 0. \end{aligned}$$

The lowest power of x is x^{m-1} . Its coefficient equated to zero gives $a_0[m(m-1) + m] = 0$.

$$m^2 = 0 \text{ as } a_0 \neq 0. \quad \therefore m = 0, 0.$$

The coefficients of $x^m, x^{m+1}, \dots$ equated to zero give

$$a_1[(m+1)m + m + 1] = 0, \text{ i.e., } a_1 = 0$$

$$a_2(m+2)^2 + a_0 = 0, a_3(m+3)^2 + a_1 = 0, a_4(m+4)^2 + a_2 = 0 \text{ and so on.}$$

Clearly $a_3 = a_5 = a_7 \dots = 0$.

Also

$$a_2 = -\frac{a_0}{(m+2)^2}, a_4 = -\frac{a_2}{(m+4)^2} = \frac{a_0}{(m+2)^2(m+4)^2} \text{ etc.}$$

$$\therefore y = a_0x^m \left[1 - \frac{x^2}{(m+2)^2} + \frac{x^4}{(m+2)^2(m+4)^2} - \frac{x^6}{(m+2)^2(m+4)^2(m+6)^2} + \dots \right]$$

Putting $m = 0$, the first solution is

$$y_1 = a_0 \left[1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots \right]$$

This gives only one solution instead of two. To get the second solution, differentiate (ii) partially w.r.t. m .

$$\frac{\partial y}{\partial m} = y \log x + a_0 x^m \left\{ \frac{x^2}{(m+2)^2} \frac{2}{m+2} - \frac{x^4}{(m+2)^2 (m+4)^2} \left[\frac{2}{m+2} + \frac{2}{m+4} \right] + \dots \right\}$$

$\therefore$ the second solution is $y_2 = \left(\frac{\partial y}{\partial m} \right)_{m=0}$

$$= y_1 \log x + a_0 \left\{ \frac{1}{2^2} x^2 - \frac{1}{2^2 \cdot 4^2} \left(1 + \frac{1}{2} \right) x^4 + \frac{1}{2^2 \cdot 4^2 \cdot 6^2} \left(1 + \frac{1}{2} + \frac{1}{3} \right) x^6 - \dots \right\}$$

Hence the complete solution is $y = c_1 y_1 + c_2 y_2$.

$$y = (C_1 + C_2 \log x) \left[1 - \frac{1}{2^2} x^2 + \frac{1}{2^2 \cdot 4^2} x^4 - \frac{1}{2^2 \cdot 4^2 \cdot 6^2} x^6 + \dots \right]$$
$$+ C_2 \left\{ \frac{1}{2^2} x^2 - \frac{1}{2^2 \cdot 4^2} \left(1 + \frac{1}{2} \right) x^4 + \frac{1}{2^2 \cdot 4^2 \cdot 6^2} \left(1 + \frac{1}{2} + \frac{1}{3} \right) x^6 - \dots \right\}$$

where $C_1 = a_0 c_1$, $C_2 = a_0 c_2$.

Case III. When roots of indicial equation are distinct and differ by an integer, making a coefficient of y infinite.

Let m_1 and m_2 be the roots such that $m_1 < m_2$. If some of the coefficients of y series become infinite when $m = m_1$, we modify the form of y by replacing a_0 by $b_0 (m - m_1)$. Then the complete solution is

$$y = C_1 (y)_{m_2} + C_2 \left(\frac{\partial y}{\partial m} \right)_{m_1}$$

Example

Obtain the series solution of the equation

$$x(1-x) \frac{d^2 y}{dx^2} - (1+3x) \frac{dy}{dx} - y = 0.$$

Solution. Here $x = 0$ is a singular point, since coefficient of y'' is zero at $x = 0$.

$\therefore$ substituting

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots$$

$$\frac{dy}{dx} = m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots$$

$$\frac{d^2 y}{dx^2} = m(m-1) a_0 x^{m-2} + (m+1)m a_1 x^{m-1} + (m+2)(m+1) a_2 x^m + \dots$$

in the given equation, we obtain

$$x(1-x)[m(m-1)a_0x^{m-2} + (m+1)ma_1x^{m-1} + (m+2)(m+1)a_2x^m + \dots] \\ - (1+3x)[ma_0x^{m-1} + (m+1)a_1x^m + (m+2)a_2x^{m+1} + \dots] - [a_0x^m + a_1x^{m+1} + a_2x^{m+2} + \dots] = 0$$

Equating to zero the coefficients of the lowest power of x , we get $a_0[m(m-1) - m] = 0$, ($a_0 \neq 0$),
 $m(m-2) = 0$, i.e. $m = 0, 2$ i.e., the two roots are distinct and differ by an integer.

Equating to zero the coefficients of successive powers of x , we get

$$(m-1)a_1 = (m+1)a_0, ma_2 = (m+2)a_1, (m+1)a_3 = (m+3)a_2 \text{ and so on.}$$

$$a_1 = \frac{m+1}{m-1}a_0, a_2 = \frac{(m+1)(m+2)}{(m-1)m}a_0, a_3 = \frac{(m+1)(m+2)(m+3)}{(m-1)m(m+1)}a_0 \text{ etc.}$$

Thus (i) becomes

$$y = a_0x^m \left[1 + \frac{m+1}{m-1}x + \frac{(m+1)(m+2)}{(m-1)m}x^2 + \frac{(m+1)(m+2)(m+3)}{(m-1)m(m+1)}x^3 + \dots \right]$$

Putting $m = 2$ (greater of the two roots) in (ii), the first solution is

$$y_1 = a_0x^2 \left[1 + 3x + \frac{3.4}{2}x^2 + \frac{4.5}{2}x^3 + \dots \right]$$

If we put $m = 0$ in (ii), the coefficients become infinite.

To obviate this difficulty, put $a_0 = b_0 (m - 0)$ so that

$$y = b_0 x^m \left[m + \frac{m(m+1)}{m-1} x + \frac{(m+1)(m+2)}{m-1} x^2 + \frac{(m+1)(m+2)(m+3)}{(m-1)(m+1)} x^3 + \dots \right]$$
$$\frac{\partial y}{\partial m} = b_0 x^m \log x \left[m + \frac{m(m+1)}{m-1} x + \frac{(m+1)(m+2)}{m-1} x^2 + \frac{(m+1)(m+2)(m+3)}{(m-1)(m+1)} x^3 + \dots \right]$$
$$+ b_0 x^m \left[1 + \frac{m^2 - 2m - 1}{(m-1)^2} x + \frac{m^2 - m - 5}{(m-1)^2} x^2 + \frac{m^2 - 2m - 11}{(m-1)^2} x^3 + \dots \right]$$

$$\begin{aligned}\therefore \text{ the second solution is } y_2 &= \left(\frac{\partial y}{\partial m} \right)_{m=0} \\ &= b_0 \log x [-1.2x^2 - 2.3x^3 - 3.4x^4 - \dots] + b_0[1 - x - 5x^2 - 11x^3 - \dots]\end{aligned}$$

Hence the complete solution is $y = c_1 y_1 + c_2 y_2$

$$\begin{aligned}y &= \frac{1}{2} c_1 a_0 [1.2x^2 + 2.3x^3 + 3.4x^4 + \dots] - b_0 c_2 \log x [1.2x^2 + 2.3x^3 + 3.4x^4 + \dots] \\ &\quad - b_0 c_2 [-1 + x + 5x^2 + 11x^3 + \dots] \\ y &= (C_1 + C_2 \log x) (1.2x^2 + 2.3x^3 + 3.4x^4 + \dots) + C_2 (-1 + x + 5x^2 + 11x^3 + \dots) \\ &\quad \text{where } C_1 = \frac{1}{2} c_1 a_0, C_2 = -b_0 c_2\end{aligned}$$

Bessel's equation

$$x^2 y'' + xy' + (x^2 - n^2)y = 0$$

Let the series solution (1) be

$$y = \sum_{m=0}^{\infty} c_m x^{k+m}, \quad c_0 \neq 0. \quad \dots(2)$$

$$\therefore y' = \sum_{m=0}^{\infty} c_m (k+m) x^{k+m-1} \quad \text{and} \quad y'' = \sum_{m=0}^{\infty} c_m (k+m)(k+m-1) x^{k+m-2}$$

Substitution for y, y', y'' in (1) now gives

$$x^2 \sum_{m=0}^{\infty} c_m (k+m)(k+m-1) x^{k+m-2} + x \sum_{m=0}^{\infty} c_m (k+m) x^{k+m-1} + (x^2 - n^2) \sum_{m=0}^{\infty} c_m x^{k+m} = 0$$

$$\sum_{m=0}^{\infty} c_m \{(k+m)(k+m-1) + (k+m) - n^2\} x^{k+m} + \sum_{m=0}^{\infty} c_m x^{k+m+2} = 0.$$

But the bracketed expression in the above identity

$$= (k+m)^2 - (k+m) + (k+m) - n^2 = (k+m)^2 - n^2 = (k+m+n)(k+m-n).$$

So the above identity becomes

$$\sum_{m=0}^{\infty} c_m (k+m+n)(k+m-n) x^{k+m} + \sum_{m=0}^{\infty} c_m x^{k+m+2} \equiv 0. \quad \dots(3)$$

Equating to zero the smallest power of x , namely x^k , (3) gives the indicial equation $c_0(k+n)(k-n) = 0$ i.e., $(k+n)(k-n) = 0$ as $c_0 \neq 0$. Its roots are $k = n, -n$.

Next equating to zero the coefficient of x^{k+1} in (3) gives

$$c_1(k+1+n)(k+1-n) = 0, \quad \text{so that} \quad c_1 = 0 \quad \text{for} \quad k = n \quad \text{and} \quad k = -n.$$

Finally equating to zero the coefficient of x^{k+m} in (3) gives

$$c_m(k+m+n)(k+m-n) + c_{m-2} = 0 \quad \text{or} \quad c_m = \frac{1}{(k+m+n)(n-k-m)} c_{m-2}. \quad \dots(4)$$

Putting $m = 3, 5, 7, \dots$ in (4) and using $c_1 = 0$, we find

$$c_1 = c_3 = c_5 = c_7 = \dots = 0. \quad \dots(5)$$

Putting $m = 2, 4, 6, \dots$ in (4) gives

$$c_2 = \frac{1}{(k+2+n)(n-k-2)} c_0,$$

$$c_4 = \frac{1}{(k+4+n)(n-k-4)} c_2 = \frac{1}{(k+4+n)(n-k-4)(k+2+n)(n-k-2)} c_0$$

and so on. Putting these values in (2), we get

$$y = c_0 x^k \left[1 + \frac{x^2}{(n+k+2)(n-k-2)} + \frac{x^4}{(n+k+2)(n-k-2)(n+k+4)(n-k-4)} + \dots \right]$$

Replacing k by n and $-n$ and also replacing c_0 by a and b in the above equation gives

$$y = ax^n \left\{ 1 - \frac{x^2}{4(1+n)} + \frac{x^4}{4 \cdot 8(1+n)(2+n)} - \dots \right\} \quad \dots(6)$$

and

$$y = bx^{-n} \left\{ 1 - \frac{x^2}{4(1+n)} + \frac{x^4}{4 \cdot 8(1-n)(2-n)} - \dots \right\}. \quad \dots(7)$$

The particular solution of (1) obtained from (6) above by taking the arbitrary constant $a = 1/\{2^n \Gamma(n + 1)\}$, is called the *Bessel function of the first kind of order n*. It will be denoted by $J_n(x)$. Thus, we have

$$J_n(x) = \frac{x^n}{2^n \Gamma(n + 1)} \left[1 - \frac{x^2}{4(n + 1)} + \frac{x^4}{4 \cdot 8(n + 1)(n + 2)} - \dots \right] \quad \dots(8)$$

or

$$J_n(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n + r + 1)} \left(\frac{x}{2}\right)^{2r + n} \quad \dots(9)$$

Replacing b by $1/\{2^n \Gamma(n + 1)\}$ in (7) and proceeding as above gives

$$J_{-n}(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(-n + r + 1)} \left(\frac{x}{2}\right)^{2r - n} \quad \dots(10)$$

Thus, the general solution of Bessel equation (1) when n is not an integer is

$$y = AJ_n(x) + BJ_{-n}(x), \text{ where } A \text{ and } B \text{ are arbitrary constant.}$$

Legendre's equation

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$$

Let the series solution (1) be

$$y = \sum_{m=0}^{\infty} c_m x^{k-m}, \quad c_0 \neq 0. \quad \dots(2)$$

$$\therefore y' = \sum_{m=0}^{\infty} c_m (k-m) x^{k-m-1} \quad \text{and} \quad y'' = \sum_{m=0}^{\infty} c_m (k-m)(k-m-1) x^{k-m-2}$$

Substitution for y, y', y'' in (1) now gives

$$(1-x^2) \sum_{m=0}^{\infty} c_m (k-m)(k-m-1) x^{k-m-2} - 2x \sum_{m=0}^{\infty} c_m (k-m) x^{k-m-1} + n(n+1) \sum_{m=0}^{\infty} c_m x^{k-m} = 0$$

or $\sum_{m=0}^{\infty} c_m (k-m)(k-m-1) x^{k-m-2} - \sum_{m=0}^{\infty} c_m \{(k-m)(k-m-1) + 2(k-m) - n(n+1)\} x^{k-m} = 0 \quad \dots(3)$

$$\text{Now, } (k-m)(k-m-1) + 2(k-m) - n(n+1)$$

$$= (k-m)^2 - (k-m) + 2(k-m) - n(n+1) = (k-m)^2 - n^2 + (k-m) - n$$

$$= (k-m+n)(k-m-n) + (k-m-n) = (k-m-n)(k-m+n+1).$$

Hence (3) may be re-written as

$$\sum_{m=0}^{\infty} c_m (k-m) (k-m-1) x^{k-m-2} - \sum_{m=0}^{\infty} c_m (k-m-n) (k-m+n+1) x^{k-m} = 0. \quad \dots(4)$$

(4) is an identity. To get the indicial equation, we equate to zero the coefficient of the highest power of x , namely x^k in (4) and obtain

$$c_0(k-n)(k+n+1) = 0 \quad \text{or} \quad (k-n)(k+n+1) = 0, \quad \text{as } c_0 \neq 0. \quad \dots(5)$$

So the roots of (5) are $k = n, -(n+1)$. They are unequal and differ by an integer. The next lower power of x is $k-1$. So we equate to zero the coefficient of x^{k-1} in (4) and obtain

$$c_1(k-1-n)(k+n) = 0. \quad \dots(6)$$

For $k = n$ and $-(n+1)$, neither $(k-1-n)$ nor $(k+n)$ is zero. So from (6), $c_1 = 0$. Finally, equating to zero the coefficient of x^{k-m} in (4), we have

$$c_{m-2}(k-m+2)(k-m+1) - c_m(k-m-n)(k-m+n+1) = 0$$

$$c_m = \frac{(k-m+2)(k-m+1)}{(k-m-n)(k-m+n+1)} c_{m-2}. \quad \dots(7)$$

Putting $m = 3, 5, 7, \dots$ in (7) and noting that $c_1 = 0$, we have

$$c_1 = c_3 = c_5 = c_7 = \dots = 0, \quad \dots(8)$$

which hold good for both $k = n$ and $k = -(n + 1)$.

To obtain $c_2, c_4, c_6 \dots$ etc., we consider two cases.

Case I. When $k = n$. Then, (7) becomes

$$c_m = - \frac{(n - m + 2)(n - m + 1)}{m(2n - m + 1)} c_{m-2}. \quad \dots(9)$$

Putting $m = 2, 4, 6 \dots$ in (9), we have

$$c_2 = -\frac{n(n-1)}{2(2n-1)}c_0, \quad c_4 = -\frac{(n-2)(n-3)}{4(2n-3)}c_2 = \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4 \cdot (2n-1)(2n-3)}c_0$$

and so on. Re-writting (2), we have for $k = n$

$$y = c_0 x^n + c_1 x^{n-1} + c_2 x^{n-2} + c_3 x^{n-3} + c_4 x^{n-4} + \dots \quad \dots(10)$$

Using (8) and the above values of c_2, c_4, c_6 . etc., (10) becomes (after replacing c_0 by a)

$$y = a \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4 \cdot (2n-1)(2n-3)} x^{n-4} - \dots \right]. \quad \dots(11)$$

Case II. When $k = -(n+1)$. Then, (7) becomes $c_m = \frac{(n+m-1)(n+m)}{m(2n+m+1)} c_{m-2}$. $\dots (12)$

Putting $m = 2, 4, 6, \dots$ in (12), we have

$$c_2 = \frac{(n+1)(n+2)}{2(2n+3)} c_0, \quad c_4 = \frac{(n+3)(n+4)}{4(2n+5)} c_2 = \frac{(n+1)(n+2)(n+3)(n+4)}{2 \cdot 4 \cdot (2n+3)(2n+5)} c_0$$

and so on. For $k = -(n+1)$, (2) gives

$$y = c_0 x^{-n-1} + c_1 x^{-n-2} + c_2 x^{-n-3} + c_3 x^{-n-4} + c_4 x^{-n-5} + \dots \quad \dots(13)$$

Using (8) and the above values of c_2, c_4, c_6etc., (13) becomes (after replacing c_0 by b)

Using (8) and the above values of c_2, c_4, c_6etc., (13) becomes (after replacing c_0 by b)

$$y = b \left[x^{-n-1} + \frac{(n+1)(n+2)}{2(2n+3)} x^{-n-3} + \frac{(n+1)(n+2)(n+3)(n+4)}{2 \cdot 4 \cdot (2n+3)(2n+5)} x^{-n-5} + \dots \right]. \quad \dots(14)$$

STRUM-LIOUVILLE PROBLEM

Legendre's equation $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$... (i)
 can be written as, $[(1 - x^2)y']' + \lambda y = 0$ $[\lambda = n(n + 1)]$

Bessel's equation $X^2 \frac{d^2 y}{dX^2} + X \frac{dy}{dX} + (X^2 - n^2)y = 0$ can be transformed by putting $X = kx$ (so that
 $\frac{dy}{dX} = \frac{dy}{dx} \cdot \frac{dx}{dX} = \frac{y'}{k}$, $\frac{d^2 y}{dX^2} = \frac{y''}{k^2}$ to the form

or $x^2 y'' + xy' + (k^2 x^2 - n^2)y = 0$
 or $(xy'' + y') + (\lambda x - n^2/x)y = 0$ $[\lambda = k^2]$
 or $(xy')' + (\lambda x - n^2/x)y = 0$... (ii)

Both the equations (i) and (ii) are of the form

$$[r(x)y']' + [\lambda p(x) + q(x)]y = 0 \quad \dots (1)$$

which is known as the *Sturm-Liouville equation*. Similarly Laguerre's, Hermite's equations etc. can also be reduced to (1). Thus all the above equations of engineering utility can be considered with a common approach by means of Sturm-Liouville's equation.

Equation (1) considered on some interval $a \leq x \leq b$, satisfying the conditions

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0, \beta_1 y(b) + \beta_2 y'(b) = 0 \quad \dots(2)$$

with the real constants : α_1, α_2 not both zero and β_1, β_2 not both zero. The conditions (2) at the end points are called *boundary condtions*.

A differential equation together with the boundary conditions, is called a **boundary value problem**. Equation (1) together with boundary conditions (2) is called a **Strum-Liouville problem**.

Obviously $y = 0$ is a solution of the problem for any value of the parameter λ which is a trivial solution and as such is of no practical utility. Any other solution of (1) satisfying (2) is called an *eigen function* of the problem and the corresponding value of λ is called an *eign value* of the problem.

A special case. Taking $r = p = 1$ and $q = 0$ in (1), we get

$$y'' + \lambda y = 0 \quad \dots(3)$$

Also if $\alpha_1 = \beta_1 = 1$ and $\alpha_2 = \beta_2 = 0$, then the boundary conditions (2) become

$$y(a) = 0, \quad y(b) = 0 \quad \dots(4)$$

Thus (3) and (4) constitute the *simplest form of Sturm-Liouville problem*.